BUS RESERVATION SYSTEM

Capstone Project Design Document

Problem Statement: PS-045 – Bus Reservation with Boarding Points, Seat Types and Partial Cancellation

Team ID: T062

Domain: Travel – Inter-city Bus Reservation

Technology: Java 17+ | CSV | Command-line execution

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Prepared against the supplied 10 SECONDS sample-data README and the project's 38-rule register.

1. Introduction

The project is a command-driven bus reservation engine. It loads route, stop, seat-layout, bus, trip and partner master data from CSV files; processes reservation commands in physical file order; and creates ticket, command-result, trip-chart and partner-settlement outputs.

2. Objectives

1. Maintain ordered routes with multiple boarding and dropping points.
2. Allocate seats by route-stage intervals, allowing sequential use of the same seat.
3. Support seater, sleeper and single-berth seat types, ladies-only seats and adjacent-berth restrictions.
4. Calculate stage fare, seat premium, dynamic occupancy uplift, child fare and GST.
5. Process partner and counter bookings, passenger-level cancellations and refunds.
6. Support one-time rescheduling and eligible boarding-point changes.
7. Handle operator cancellation and partner settlement.
8. Generate the evaluation CSV formats specified in the supplied README.

3. System inputs and outputs

Inputs: stops.csv, route_stops.csv, layout_seats.csv, buses.csv, trips.csv, partners.csv and commands.txt. Outputs: tickets.csv, command_results.csv, trip_chart.csv and partner_settlement.csv. The column order is defined by the sample-data README.

4. Architecture and modules

CSV data loading constructs the domain objects. Route and trip models hold ordered stops, stage fares, bus layout and departure time. Booking and seat validation checks journey stops, passenger data, seat existence, stage overlap, ladies-only and adjacent-berth constraints. Fare calculation computes stage fare, occupancy uplift, seat premium, child factor, taxable fare and GST. Cancellation, rescheduling, boarding change and reporting are implemented in the active command engine in Main.java.

5. Seat inventory model

Stops are ordered by stop_seq. A journey from stop index i to stop index j occupies the half-open interval [i, j). Two journeys on the same trip and seat overlap when newStart < existingEnd and existingStart < newEnd. Therefore A–C and C–E can use the same seat sequentially.

6. Fare calculation model

The engine sums stage fares between boarding and dropping points. Dynamic uplift applies to stage fare only. Seat premium is added, child fare factor is applied where relevant, and GST follows the bus AC flag. Amounts are rounded to paise per passenger line.

7. Cancellation and rescheduling

Cancellation uses time remaining until each passenger’s boarding-point departure: 90% for more than 24 hours; 75% from 12 hours up to and including 24 hours; 50% from 4 hours to less than 12 hours; and 0% within 4 hours. A free child cannot remain without an adult. Rescheduling is allowed once, at least six hours before original boarding and within the permitted 30-day range. The fee is Rs 50 per active passenger plus any positive fare difference; lower fares do not produce a refund.

8. Business-rule register

The authoritative detailed register is docs/business-rules.md and contains BR-01 through BR-38. Related rules may be grouped into functional topics in this design document; the groups do not replace the 38 individual rule IDs.

10. Testing and verification

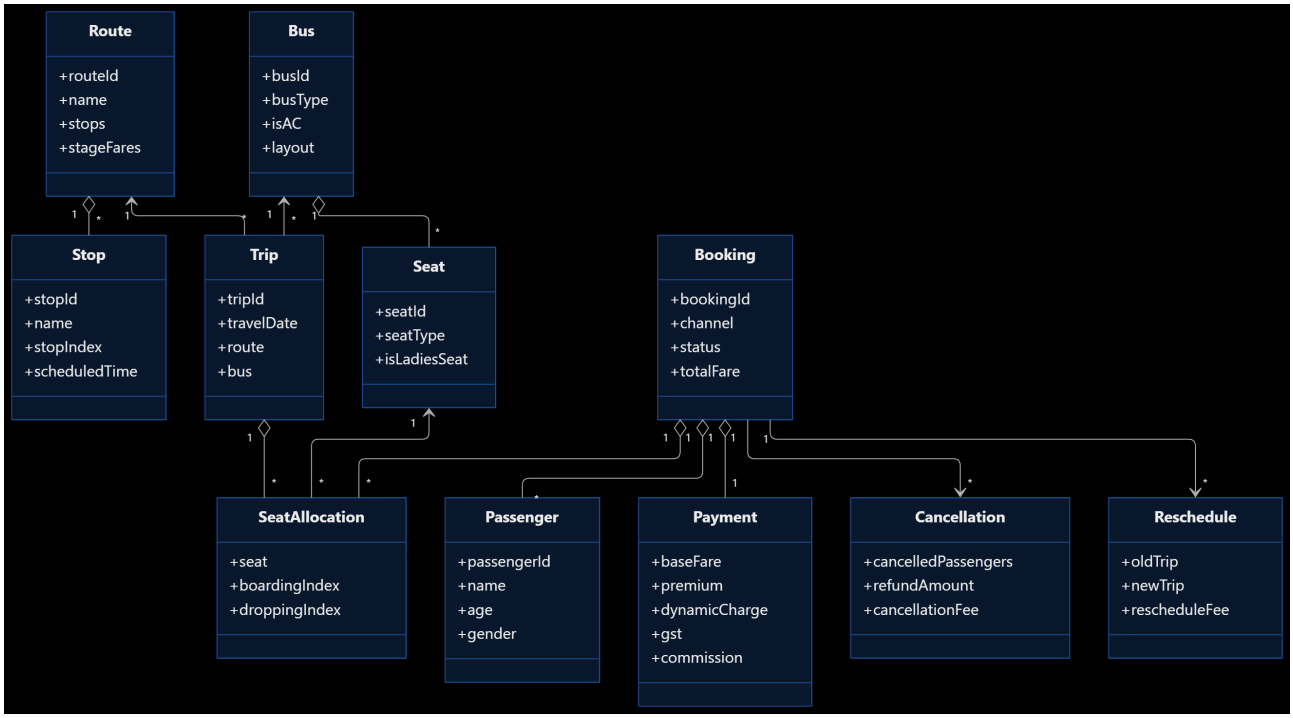
The executable regression suite is tests/verify_project.py. It runs the official 127-command sample and additional interaction scenarios. The latest run is recorded in docs/verification-report.md. The sample pack does not include expected output CSVs, and final evaluation uses hidden input; passing visible tests is not a guarantee of hidden-set correctness.

11. Open clarification items

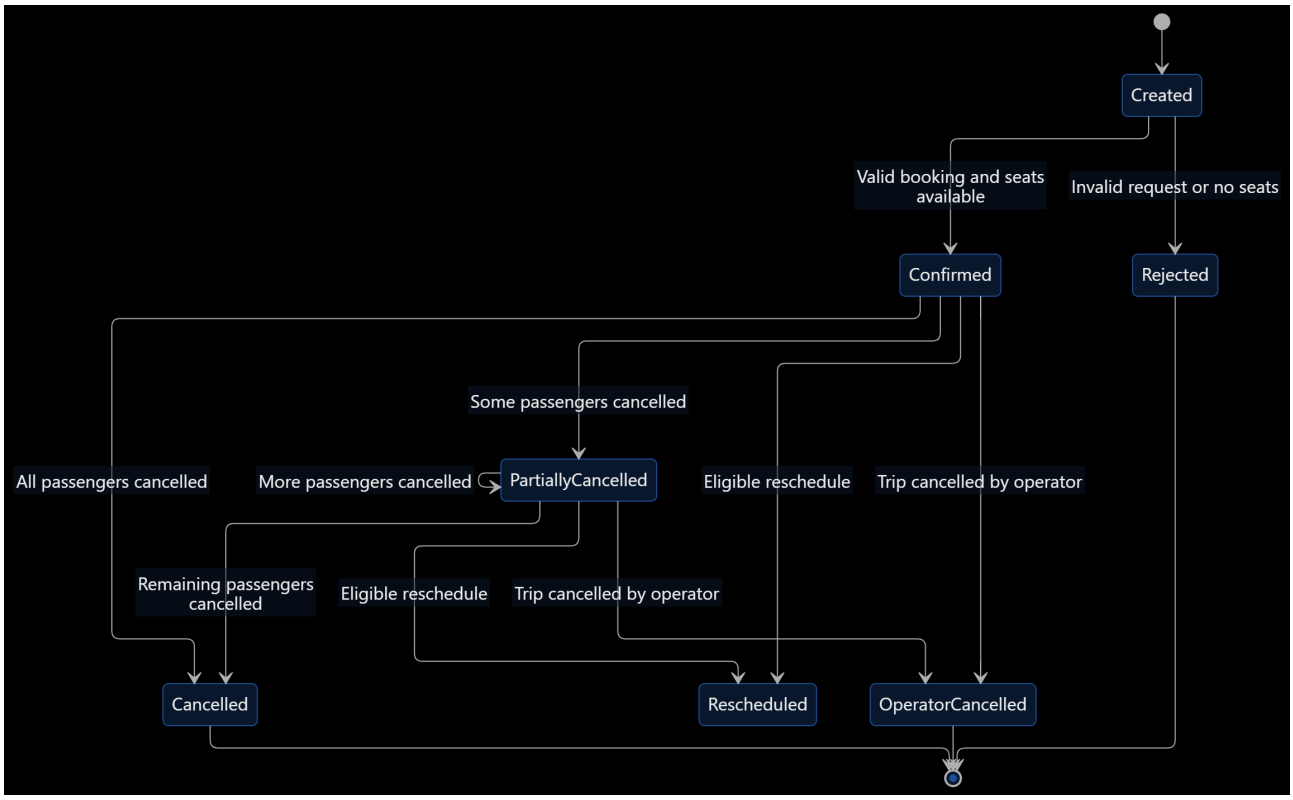
Implementation assumptions and five mentor clarification topics are recorded in docs/decision_log.md. They should be updated if the assignment thread provides authoritative answers.

9. Design diagrams

Class diagram



Booking lifecycle state diagram


Created
Valid booking and seats available
Invalid request or no seats
Confirmed
Rejected
Some passengers cancelled
All passengers cancelled
More passengers cancelled
PartiallyCancelled
Eligible reschedule
Trip cancelled by operator
Remaining passengers cancelled
Eligible reschedule
Trip cancelled by operator
Cancelled
Rescheduled
OperatorCancelled